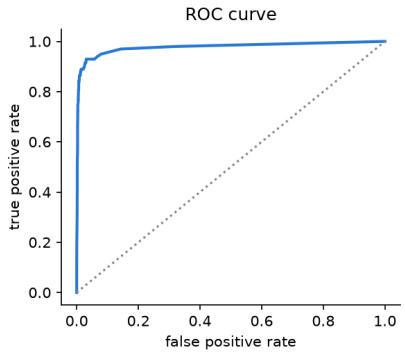
Model Performance Report

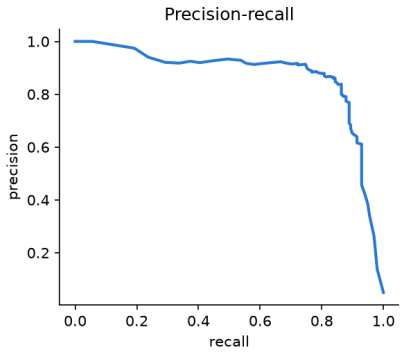
Model gbm-20261110-0237-3afcd506 · generated November 09, 2026

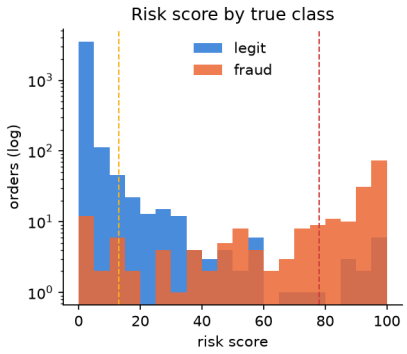
Summary

Model **gbm-20261110-0237-3afcd506**, a gradient-boosted tree ensemble over 22 engineered signals, trained on 15,960 orders and tested on the most recent 3,990 (time-based split; test fraud rate 5.0%). ROC-AUC **0.9759**, PR-AUC **0.8697** (logistic-regression baseline PR-AUC 0.8385).

Operating point	Precision	Recall	Orders flagged	Confusion (TN / FP / FN / TP)
Review (score ≥ 13)	61.6%	91.4%	7.4%	3679 / 113 / 17 / 181
Block (score ≥ 78)	92.1%	65.1%	3.5%	3781 / 11 / 69 / 129







Threshold trade-off

Threshold	Orders flagged	Precision	Recall	Fraud $ caught
10	325 (8.1%)	56.6%	92.9%	$38,180
13	294 (7.4%)	61.6%	91.4%	$37,970
20	249 (6.2%)	70.7%	88.9%	$37,066
50	182 (4.6%)	87.9%	80.8%	$32,649
78	140 (3.5%)	92.1%	65.2%	$26,678
90	113 (2.8%)	92.9%	53.0%	$24,794

Champion vs baselines

Model	ROC-AUC	PR-AUC	Precision@13	Recall@13	FPR@13
champion (gbm-20261110-0237-3afcd506)	0.9759	0.8697	61.6%	91.4%	2.98%
logistic regression	0.9766	0.8385	19.0%	97.5%	21.76%
rules checklist	0.9476	0.7218	11.3%	96.0%	39.48%

Automated quality gates (latest QA run)

Check	Result	Status
mq-auc · model_quality	ROC-AUC 0.976 | PR-AUC 0.870	PASS
mq-errors · error_rates	FPR 2.98% | FNR 8.59% | $1,240 fraud missed	PASS
mq-signals · signal_detection	precision 99% | coverage 97% over 181 caught fraud orders	PASS
mq-logic · scoring_logic	monotonic 100% | bounds ok | deterministic yes	PASS
mq-drift · drift	score PSI 0.010 | worst feature account_age_days PSI 0.146	PASS
mq-compare · model_comparison	champion (gbm-20261110-0237-3afcd506): PR-AUC 0.870 | logistic regression: PR-AUC 0.839 | rules checklist: PR-AUC 0.722	PASS
mq-calibration · calibration	ECE 0.0077 | Brier 0.0129	PASS
mq-reliance · feature_reliance	top feature checkout_seconds 26%	PASS

Drift watch-list (PSI, training vs current)

Feature	PSI	Status
account_age_days	0.146	moderate
amount_to_baseline	0.0309	stable
users_per_card	0.0083	stable
account_age_known	0.0077	stable
login_attempts	0.0066	stable
failed_logins	0.0062	stable

Limitations

- Trained on synthetic data; real-store performance must be re-measured on labelled history.
- Cross-order signals (shared devices / cards) are computed over all known history, so early orders in a ring look cleaner than later ones.
- Labels arrive late (chargebacks take weeks): monitor drift weekly and retrain monthly using reviewer decisions.